

Part I

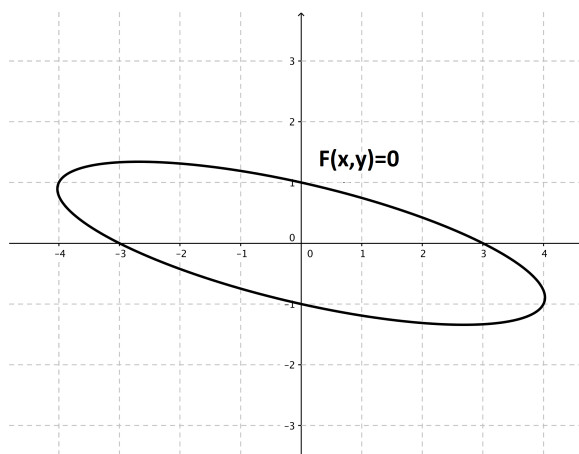
Recitation Questions

Problem 1

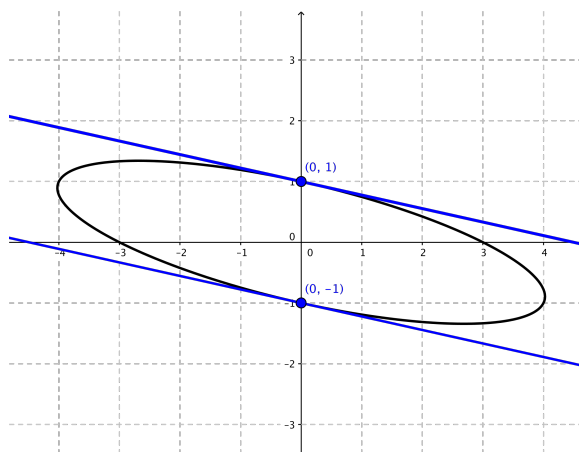
February 21, 2026

Problem 1

Problem 1 On the graph below, sketch the tangent lines at $x = 0$. Then, explain why both the x -coordinate and the y -coordinate are generally needed to find the slope of the tangent line at a point on the graph of an equation of the form $F(x, y) = 0$



Explanation. Given the equation $F(x, y) = 0$, its graph may not pass the “vertical line test”. As illustrated in the figure below, fixing a value for x (in this case $x = 0$) will not typically specify a unique point on the graph because there may be more than one corresponding y -value. That is why you need to specify both the x and the y coordinate, to make sure that you are giving the coordinates for a unique point on the graph. In this case, there are two tangent lines to the curve at $x = 0$. One passes through the point $(0, 1)$ and the other through $(0, -1)$.

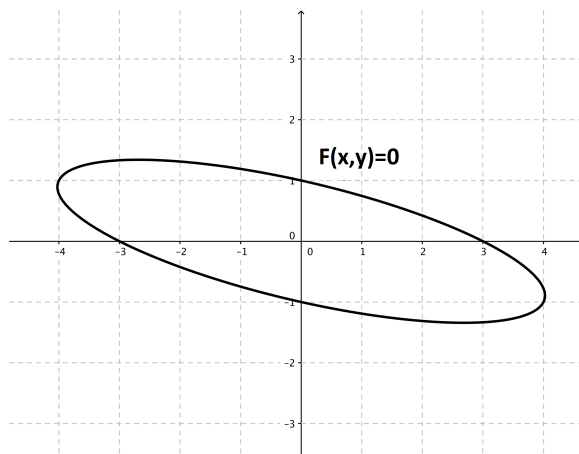


Problem 2

February 21, 2026

Problem 2

Problem 2 Consider the equation $x^2 + 4xy + 9y^2 = 9$. Note: This equation is equivalent to $x^2 + 4xy + 9y^2 - 9 = 0$. Therefore it has a form $F(x, y) = 0$



- (a) Find $\frac{dy}{dx}$.

Explanation.

$$\begin{aligned}\frac{d}{dx}(x^2 + 4xy + 9y^2) &= \frac{d}{dx}(9) \\ 2x + \left(4y + 4x \frac{dy}{dx}\right) + 18y \frac{dy}{dx} &= 0 \\ 4x \frac{dy}{dx} + 18y \frac{dy}{dx} &= -2x - 4y \\ (4x + 18y) \frac{dy}{dx} &= -2x - 4y \\ \frac{dy}{dx} &= \frac{-2x - 4y}{4x + 18y}\end{aligned}$$

provided that $4x + 18y \neq 0$.

- (b) Find the equation(s) of the tangent line(s) when $x = 0$. Draw the tangent line(s) on the above picture.

Explanation. Plugging into the original equation, we see that when $x = 0$ we have that

$$0^2 + 4(0)y + 9y^2 = 9 \quad \implies \quad y^2 = 1 \quad \implies \quad y = \pm 1$$

Then,

$$\left[\frac{dy}{dx}\right]_{(0,1)} = \frac{-4}{18} = -\frac{2}{9}$$

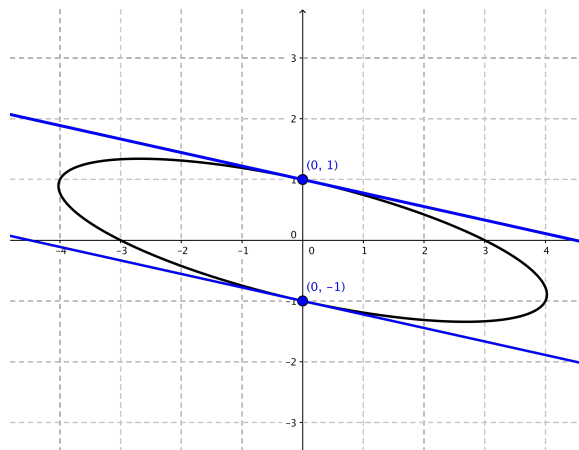
Problem 2

$$\left[\frac{dy}{dx} \right]_{(0,-1)} = \frac{4}{-18} = -\frac{2}{9}$$

So there are two tangent lines to the graph of the given equation when $x = 0$. The two points are $(0, 1)$ and $(0, -1)$, and the tangent lines at both points have slope $-\frac{2}{9}$. Thus, the equations of the two tangent lines are

$$y - 1 = -\frac{2}{9}(x - 0) \quad \implies \quad y = -\frac{2}{9}x + 1$$

$$y + 1 = -\frac{2}{9}(x - 0) \quad \implies \quad y = -\frac{2}{9}x - 1$$



- (c) Find the point(s) where the tangent line is horizontal. Draw the point(s) and line(s) on the above picture.

Explanation. A line is horizontal if and only if its slope is 0. So we are looking for the points (x_0, y_0) such that $\left[\frac{dy}{dx} \right]_{(x_0, y_0)} = 0$. So we solve:

$$\frac{-2x - 4y}{4x + 18y} = 0 \quad \implies \quad -2x - 4y = 0 \quad \implies \quad x = -2y$$

But we also need the point to lie on the the given curve. So, letting $x = -2y$, we solve:

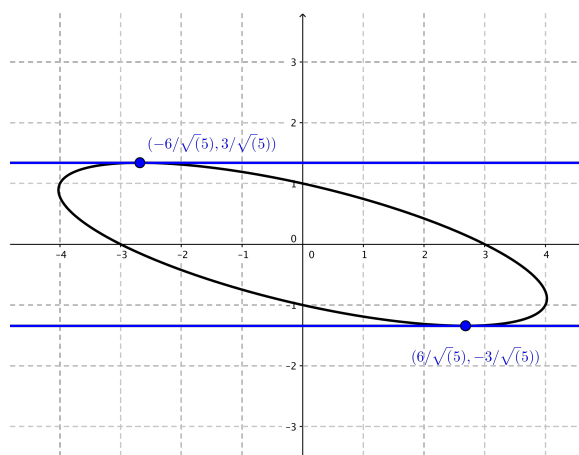
$$\begin{aligned} x^2 + 4xy + 9y^2 &= 9 \\ (-2y)^2 + 4(-2y)y + 9y^2 &= 9 \\ 4y^2 - 8y^2 + 9y^2 &= 9 \\ 5y^2 &= 9 \end{aligned}$$

Problem 2

$$y^2 = \frac{9}{5}$$

$$y = \pm \frac{3}{\sqrt{5}}$$

So there exists two solutions to the given equation where the tangent line to the graph has 0 slope. Those two points are $\left(-\frac{6}{\sqrt{5}}, \frac{3}{\sqrt{5}}\right)$ and $\left(\frac{6}{\sqrt{5}}, -\frac{3}{\sqrt{5}}\right)$.

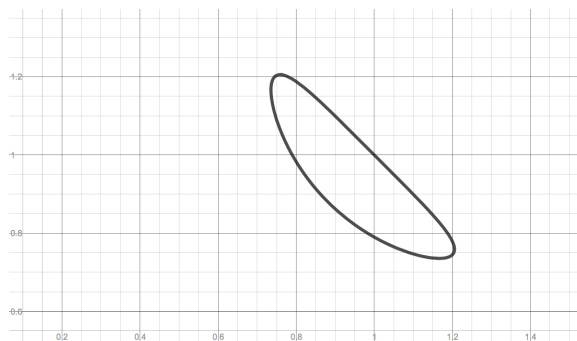


Problem 3

February 21, 2026

Problem 3

Problem 3 A part of the curve with equation $\cos(\pi xy) + x + y = 1$ is sketched below.



- (a) Use the implicit differentiation to find the derivative dy/dx .

Explanation. Implicitly differentiate equation:

$$\frac{d}{dx}(\cos(\pi xy) + x + y) = \frac{d}{dx}(1) \implies -\sin(\pi xy) \cdot (\pi y + \pi x \frac{dy}{dx}) + 1 + \frac{dy}{dx} = 0$$

Solve for $\frac{dy}{dx}$:

$$\begin{aligned} -\sin(\pi xy) \cdot (\pi y + \pi x \frac{dy}{dx}) + 1 + \frac{dy}{dx} &= 0 \implies -\pi y \sin(\pi xy) - \pi x \frac{dy}{dx} \sin(\pi xy) + 1 + \frac{dy}{dx} = 0 \\ &\implies (1 - \pi x \sin(\pi xy)) \frac{dy}{dx} = \pi y \sin(\pi xy) - 1 \\ &\implies \frac{dy}{dx} = \frac{\pi y \sin(\pi xy) - 1}{1 - \pi x \sin(\pi xy)} \quad \text{if } 1 - \pi x \sin(\pi xy) \neq 0 \end{aligned}$$

- (b) Consider the point $(1, 1)$. Show (algebraically) that this point lies on the curve.

Explanation. $(1, 1)$ lies on this curve:

$$\begin{aligned} \cos(\pi \cdot 1 \cdot 1) + 1 + 1 &= \cos(\pi) + 2 \\ &= -1 + 2 = 1 \end{aligned}$$

- (c) Find the equation of the line tangent to the curve at $(1, 1)$. Draw this line in the figure above.

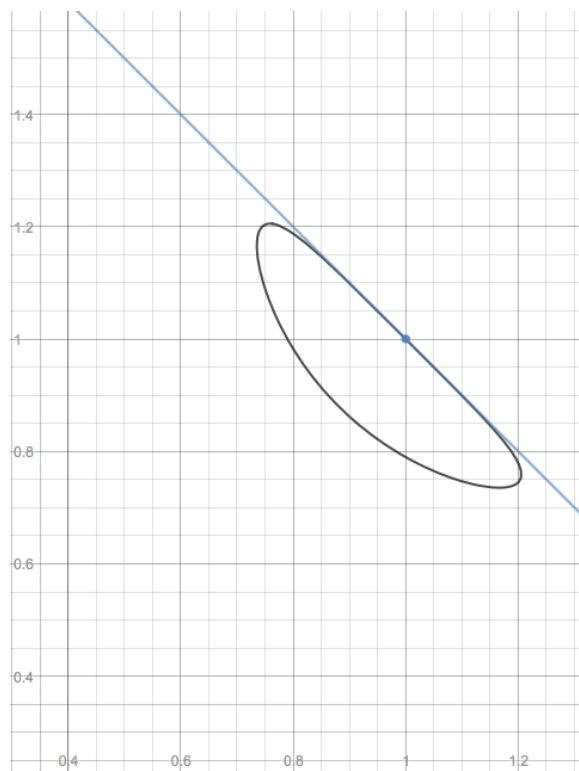
Explanation. Slope of tangent line:

$$\begin{aligned} \left[\frac{dy}{dx} \right]_{(1,1)} &= \frac{\pi \cdot (1) \cdot \sin(\pi(1) \cdot 1) - 1}{1 - \pi \cdot (1) \cdot \sin(\pi(1) \cdot (1))} \\ &= \frac{\pi \sin(\pi) - 1}{1 - \pi \sin(\pi)} \\ &= \frac{-1}{1} = -1 \end{aligned}$$

Equation of tangent line:

$$y - 1 = -1(x - 1) \implies y = -x + 2$$

Graph of tangent line:



Problem 4

February 21, 2026

Problem 4 For each of the curves given by the following equations, find a formula for the slope of the tangent line at a point (x, y) .

(a) $e^{x^2y^3} - 5x + 7y = 36$

Explanation.

$$e^{x^2y^3} \left(2xy^3 + x^2(3y^2) \frac{dy}{dx} \right) - 5 + 7 \frac{dy}{dx} = 0$$

$$2xy^3 e^{x^2y^3} + 3x^2y^2 e^{x^2y^3} \frac{dy}{dx} - 5 + 7 \frac{dy}{dx} = 0$$

$$3x^2y^2 e^{x^2y^3} \frac{dy}{dx} + 7 \frac{dy}{dx} = -2xy^3 e^{x^2y^3} + 5$$

$$\left(3x^2y^2 e^{x^2y^3} + 7 \right) \frac{dy}{dx} = -2xy^3 e^{x^2y^3} + 5$$

$$\frac{dy}{dx} = \frac{-2xy^3 e^{x^2y^3} + 5}{3x^2y^2 e^{x^2y^3} + 7}$$

(b) $7 = 22 \tan(y) + \frac{4}{x} - \frac{7}{y}$

Explanation.

$$0 = 22 \sec^2(y) \frac{dy}{dx} - \frac{4}{x^2} + \frac{7}{y^2} \frac{dy}{dx}$$

$$22 \sec^2(y) \frac{dy}{dx} + \frac{7}{y^2} \frac{dy}{dx} = \frac{4}{x^2}$$

$$\frac{dy}{dx} = \frac{\frac{4}{x^2}}{22 \sec^2(y) + \frac{7}{y^2}}$$

(c) $\cos(xy) - x^3 = 5y^3$

Explanation.

$$-\sin(xy) \cdot \left(y + x \frac{dy}{dx} \right) - 3x^2 = 15y^2 \frac{dy}{dx}$$

$$-y \sin(xy) - x \sin(xy) \frac{dy}{dx} - 3x^2 = 15y^2 \frac{dy}{dx}$$

$$x \sin(xy) \frac{dy}{dx} + 15y^2 \frac{dy}{dx} = -y \sin(xy) - 3x^2$$

$$\frac{dy}{dx} = \frac{-y \sin(xy) - 3x^2}{x \sin(xy) + 15y^2}$$

Provided that $x \sin(xy) + 15y^2 \neq 0$. It is worth pointing out that equivalent condition was not necessary for parts (a) and (b) because the denominator of those solutions cannot be 0.

Problem 5

February 21, 2026

Problem 5 The volume of a doughnut with an inner radius of a and an outer radius of b is

$$V = \pi^2 \frac{(b+a)(b-a)^2}{4}.$$

Find db/da if the volume of a doughnut is $64\pi^2$ and does not change.

Explanation. We have to (implicitly) differentiate

$$64\pi^2 = \pi^2 \frac{(b+a)(b-a)^2}{4}.$$

Before doing this we'll perform a bit of algebra to simplify our calculations:

$$64\pi^2 = \pi^2 \frac{(b+a)(b-a)^2}{4} \implies 256 = (b+a)(b-a)^2.$$

Now we'll differentiate with respect to a :

$$\begin{aligned} 256 = (b+a)(b-a)^2 &\implies 0 = \left(\frac{db}{da} + 1\right)(b-a)^2 + (b+a)2 \cdot (b-a) \cdot \left(\frac{db}{da} - 1\right) \\ &\implies 0 = \frac{db}{da}(b-a)^2 + (b-a)^2 + 2(b^2 - a^2)\frac{db}{da} - 2(b^2 - a^2). \end{aligned}$$

To finish we solve for $\frac{db}{da}$:

$$\begin{aligned} 0 &= \frac{db}{da}(b-a)^2 + (b-a)^2 + 2(b^2 - a^2)\frac{db}{da} - 2(b^2 - a^2) \\ &\implies -\left(\frac{db}{da}(b-a)^2 + 2(b^2 - a^2)\frac{db}{da}\right) = (b-a)^2 - 2(b^2 - a^2) \\ &\implies \frac{db}{da} = \frac{(b-a)^2 - 2(b^2 - a^2)}{-((b-a)^2 + 2(b^2 - a^2))}. \end{aligned}$$

Problem 6

February 21, 2026

Problem 6

Problem 6 The curve is given by the equation $x^{1/3} + y^{2/3} = 2$. Find $\frac{d^2y}{dx^2}$.

Explanation. Take the first derivative:

$$\frac{1}{3}x^{-2/3} + \frac{2}{3}y^{-1/3} \cdot \frac{dy}{dx} = 0$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{\frac{-1}{3x^{2/3}}}{\frac{2}{3y^{1/3}}} = \frac{-y^{1/3}}{2x^{2/3}}$$

Take the second derivative:

$$\frac{d^2y}{dx^2} = \frac{-\frac{1}{3}y^{-2/3} \cdot \frac{dy}{dx} \cdot 2x^{2/3} + y^{1/3} \cdot \frac{4}{3}x^{-1/3}}{4x^{4/3}}$$

We need to substitute in for $\frac{dy}{dx}$ we have:

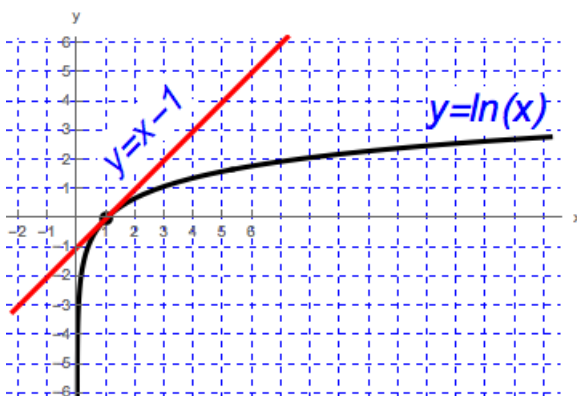
$$\frac{d^2y}{dx^2} = \frac{-\frac{1}{3}y^{-2/3} \cdot \frac{-y^{1/3}}{2x^{2/3}} \cdot 2x^{2/3} + y^{1/3} \cdot \frac{4}{3}x^{-1/3}}{4x^{4/3}}$$

Problem 7

February 21, 2026

Problem 7 Sketch both the curve $y = \ln(x)$ and the tangent line to the curve at the point where $x = 1$.

Then, write an equation of the tangent line to the curve $y = \ln(x)$ at the point where $x = 1$.



Explanation.

In order to write an equation of the tangent line we need to find coordinates of the point $P(1, \ln(x))$, and the slope of the line, $y'(1)$.

$P(1, 0)$, and, since $y'(x) = \frac{1}{x}$, we get that $y'(1) = 1$. Therefore, the equation is given by
 $y = x - 1$.

Problem 8

February 21, 2026

Problem 8 (a) Let f be a positive differentiable function, defined on an open interval I . Find the formula for the derivative of the function $\ln(f(x))$.

Explanation. $\frac{d}{dx} \ln(f(x)) = \frac{1}{f(x)} \cdot f'(x) = \frac{f'(x)}{f(x)}$

(b) Using the formula obtained in part (a), compute the derivatives of the following functions.

(i) $f(x) = \ln(x^2 + x + 1)$

Explanation. $f'(x) = \frac{2x + 1}{x^2 + x + 1}$

(ii) $f(x) = \ln(\sec(x) + \tan(x))$

Explanation. $f'(x) = \frac{\sec(x)\tan(x) + \sec^2(x)}{\sec(x) + \tan(x)} = \frac{\sec(x)(\tan(x) + \sec(x))}{\sec(x) + \tan(x)} = \sec(x)$

(iii) $f(x) = \ln(\ln(x))$

Explanation. $f'(x) = \frac{\frac{1}{x}}{\ln(x)} = \frac{1}{x \ln(x)}$

Problem 9

February 21, 2026

Problem 9 Compute $f'(x)$.

(a) $f(x) = x \ln(x)$

Explanation. $f'(x) = x \cdot \frac{1}{x} + \ln(x) = 1 + \ln(x)$

(b) $f(x) = \sin(x) \left(\ln(\sec(x) + 1) \right)$

Explanation. $f'(x) = \sin(x) \frac{\sec(x) \tan(x)}{\sec(x) + 1} + \cos(x) \left(\ln(\sec(x) + 1) \right)$

(c) $f(x) = 2^x \sqrt{\ln(5x + 7)}$

Explanation. $f'(x) = \frac{2^x}{2\sqrt{\ln(5x + 7)}} \cdot \frac{5}{5x + 7} + 2^x \ln(2) \sqrt{\ln(5x + 7)}$